

QUALITY ASSURANCE (QA) TEST PLAN

Project Name : Smart File System Automation and Management System

1. Objective The objective of this testing activity is to verify that the Smart File System Automation and Management System functions according to the requirements specified in the Software Requirements Specification (SRS) document. Testing will ensure that automatic file classification, file naming rule validation, duplicate detection, backup mechanisms, file encryption, and report generation work accurately, securely, and reliably.

File System Automation

Modules Tested

1. Classify Module
2. Validation Module
3. Password Module

Test Cases:

Test Case ID	Module	Test Scenario	Test Data	Expected Result	Status
TC001	Create File	Create a new file with valid name	project.pdf	File created successfully	Pass
TC002	Create File	Create duplicate file	project.pdf	Display "File already exists."	Pass
TC003	Create File	Empty filename	""	Display validation error	Pass
TC004	Create File	Invalid filename	file<>.txt	Display "Invalid filename."	Pass
TC005	Password	Create file without password	No	File created without password	Pass

TC006	Password	Create file with password	Yes, Password: 1234	Password stored successfully	Pass
TC007	Password	Empty password	""	Ask user to enter valid password	Pass
TC008	Rename File	Rename existing file	project.pdf → report.pdf	File renamed successfully	Pass
TC009	Rename File	Rename non-existing file	abc.pdf	Display "File not found."	Pass
TC010	Rename File	Rename to existing filename	report.pdf	Display "File already exists."	Pass
TC011	Delete File	Delete existing file	report.pdf	File deleted successfully	Pass
TC012	Delete File	Delete protected file with correct password	Password: 1234	File deleted	Pass
TC013	Delete File	Delete protected file with wrong password	Password: 0000	Display "Incorrect password."	Pass
TC014	Delete File	Delete non-existing file	xyz.txt	Display "File not found."	Pass

TC015	Display Records	Display when records exist	-	Show all file records	Pass
TC016	Display Records	Display when no records exist	-	Display "No records found."	Pass